7. QUANTITATIVE ANALYSIS

Consider an extension of the model of Section 5 where upon meeting, the firm–worker match draws the initial productivity from a cumulative distribution function with $G_0(x) = 1 - \left(\frac{\varepsilon_0}{x}\right)^{\alpha}$ if $\varepsilon_0 \leq x$ and $G_0(x) = 1 - \left(\frac{\varepsilon_0}{x}\right)^{\alpha} = 0$ otherwise, where $\varepsilon_0 > 0$. Subsequent draws are from (20). As I show in Appendix B (Lemma 3), for this case the equilibrium cumulative distribution function of active firms is given by (21). The job-creation and job-destruction conditions are given by (24) and (25), but with ε_0 replacing ε in (25).[32]

I will follow the bulk of the search and matching labour literature in assuming that the meeting rate of a vacancy is $q(\theta) = \theta^{-\eta}$, with $\eta \in (0, 1)$. To calibrate this model to the U.S., I set $\tau_f = \tau_h = \tau_e = 0$, which is meant to reflect its relatively flexible labour-market regulations. Since the model will generate endogenous separations, I also set $\delta = 0$. This leaves 10 free parameters: $r, c, \varepsilon, \eta, \beta, \alpha, \lambda, \varepsilon_0, \tau_b$, and ϕ. I normalize the time period to be a quarter and set $r = c = 0{\cdot}012$, which corresponds to an annual discount factor of 0·953. I also normalize $\varepsilon = 1$. I set $\eta = 0{\cdot}72$, which is close to the upper bound of the range reported by Petrongolo and Pissarides (2001), and let $\beta = \eta$ so that there would be no inefficiencies in the absence of policy.[33]

Several studies point out that various measures of firm size are well approximated by a Pareto distribution with a curvature parameter near 1. Axtell (2001), for example, documents this fact using the number of employees and sales as measures of firm size. The model I am calibrating is in some ways too stark: for example, each firm employs a single worker, and it does not distinguish between output and sales. But notice that the equilibrium distribution of output across active firms is Pareto with curvature parameter α. These considerations may suggest a natural strategy to calibrate α. However, since the purpose here is mainly to explore the properties of the model for reasonable parametrizations rather than to devise a definitive calibration strategy, I will report results for three values of α: 1·1, 3/2, and 3. The remaining parameters $(\lambda, \varepsilon_0, \tau_b, \phi)$ are calibrated so that in the stationary equilibrium, the model implies that (i) employment spells last 2·5 years on average; (ii) an unemployed worker finds a job with probability 0·45 per month; (iii) the replacement rate is 11%; and (iv) the fraction of hoarded employed workers is relatively small, to be precise, 0·05.[34]

For the benchmark case ($\alpha = 3/2$), the values of the calibrated parameters are $\lambda = 1{\cdot}23$, $\varepsilon_0 = 0{\cdot}6$, $\tau_b = 0{\cdot}22$, and $\phi = 1{\cdot}095$. The implied means of the distributions G and G_0 are 3 and 1·84, respectively. In the equilibrium, $R = 1{\cdot}058$, and the average wage is $\bar{w} = 2{\cdot}014$. Table 1 reports the effects of small changes in the policy parameters τ_b, τ_f, τ_e, and τ_h, on the average duration of a job, $D = [\lambda G(R)]^{-1}$, the unemployment rate, u, the three measures of TFP described in Section 6.2 (*i.e.* $A = \frac{Y}{K_e^{\gamma} N^{1-\gamma}}$, $\hat{A} = \frac{Y}{K^{\gamma} N^{1-\gamma}}$, and $\tilde{A} = \frac{Y}{K^{\gamma} E^{1-\gamma}}$), and the degree of labour hoarding. I use subscripts to index the various policy scenarios, for example, A_0 denotes the level of TFP at the baseline with no policy, and A_{τ_i} denotes the level of TFP under policy τ_i, for $i = b, f, e, h$. The first row just shows that the model hits the calibration targets: the replacement rate $\tau_b/\bar{w}$ is

32. Varying ε_0 relative to ε will help calibrate the model by allowing me to vary the job-creation condition in much the same way as the usual cost of posting vacancies does in the standard formulation of the Mortensen–Pissarides model. Recall that in my formulation, a firm's only flow cost from searching is having to pay rental c on the capital k, and c will be pinned down by the interest rate.

33. Here I am following Shimer (2005), who estimates $\eta = 0.72$ and also sets $\beta = \eta$. See Hagedorn and Manovskii (2005) for an alternative calibration strategy. In Appendix B, I report results for my own alternative calibration strategy, which instead sets β to match the labour share.

34. For observations (i) and (ii), see Shimer (2005). Observation (iii) corresponds to the average value over the first five years of unemployment expressed as a percentage of previous tax earnings for a single worker reported by the OECD Jobs Study (OECD, 1994). Most of the empirical work on labour hoarding has focused on its cyclical properties and their implications for the Solow residual-based measures of technology shocks, and for the extent to which technology shocks account for the volatility of aggregate output (*e.g.* Burnside, Eichenbaum, and Rebelo, 1993). I have not been able to find empirical estimates of the average *level* of labour hoarding in my perusal of the literature. Consequently, target (iv) is not based on any hard data, but is just meant to be illustrative.

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TABLE 1

Baseline results

		$\tau_i/\bar{w}$	D	u	A_{τ_i}/A_0	$\hat{A}_{\tau_i}/\hat{A}_0$	$\tilde{A}_{\tau_i}/\tilde{A}_0$	$H(\phi)$
Baseline		0·11	10	0·069	1·000	1·000	1·000	0·050
τ_b	0·25	0·12	7	0·096	1·023	0·847	0·787	0·017
	0·19	0·10	18	0·038	0·975	1·318	1·497	0·086
τ_f	0·02	0·01	16	0·041	0·977	1·278	1·432	0·082
	0·04	0·02	54	0·012	0·954	1·973	2·717	0·114
τ_e	0·02	0·01	14	0·048	0·983	1·183	1·277	0·074
	0·04	0·02	23	0·030	0·968	1·463	1·745	0·095
τ_h	0·02	0·01	7	0·096	1·023	0·840	0·777	0·017
	0·03	0·02	6	0·108	1·034	0·786	0·706	0·001

exactly 11%, and on average a job lasts 10 quarters, which, together with a probability of 0·45 per month that an unemployed worker finds a job, implies an unemployment rate of 6·9%. Also, $H(\phi) = 0{\cdot}05$, indicating that in the equilibrium 5% of the firms hoard labour.

Increasing the replacement rate from 11% to 12% (which is achieved by increasing τ_b from 0·22 to 0·25) shifts the job-destruction condition upward as in the top-right panel of Figure 1. The increase in R causes the expected duration of a job to fall to seven quarters, and this, together with the reduction in job creation induced by the fall in θ, causes the unemployment rate to increase to 9·6%. According to the first measurement convention, A, for which the capital stock in the aggregate production function is K_e and hours worked are measured accurately, the 1% increase in the replacement rate causes a 2·3% increase in R and hence a 2·3% increase in TFP.

The measurement issues discussed in Section 6.2, however, are critical. For example, under the second measurement convention, $\hat{A}$, for which the capital stock in the aggregate production function is K and hours worked are measured accurately, the 1% increase in replacement rate causes an 18% *reduction* in the measured residual. This method is picking up such a large reduction in TFP because after the increase in unemployment benefits, the unemployment rate is significantly higher, and hence a significantly higher fraction of the capital stock that is counted as an aggregate input is now sitting idle in unmatched firms. Recall that the third measurement convention, $\tilde{A}$, is the one that uses K and employment (instead of hours worked) as inputs. According to this measure, TFP falls by 27%.[35]

Intuitively, the second way to measure inputs perfectly measures the utilization rate of labour, but does not measure the utilization rate of capital at all (it cannot even distinguish machines sitting in empty job slots from those matched to workers). And the third way does not adjust for the utilization rate of either factor. In terms of labour, it counts both workers who are employed and supplying hours as well as those who are employed but not supplying hours. The policy causes unemployment to rise, which reduces the economy-wide utilization rate of capital and labour. Thus, the measures that do not adjust for utilization capture a sharp reduction in the TFP residual.

Introducing a firing tax that corresponds to 1% of the average quarterly wage reduces unemployment by 2·8 percentage points. TFP as measured by A drops by 2·3% but rises when measured by $\hat{A}$ or $\tilde{A}$, by 28% and 43%, respectively.[36] The intuition for the first measure is clear: the productivity of the average active firm is lower because the policy makes separations more

35. The decline is larger for the third measure because the total number of hours worked, averaged over all employed workers, falls in response to the increase in τ_b.

36. In a recent paper, Chari, Restuccia, and Urrutia (2005) develop a matching model with dynamic labour contracts in which firms provide general training to workers who cannot commit to staying with the firm. Within this set-up, they find that firing costs can increase measured productivity, and they argue that this result is borne out by European data.

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costly, and hence lower-quality matches survive in the presence of firing restrictions. The large increases in aggregate productivity as measured by $\hat{A}$ and $\tilde{A}$ are again due to the fact that the unemployment rate is lower, and hence the utilization rate of the factors is higher with the policy. The magnitudes of the effects of the remaining policies are similar.[37]

Tables 2 and 3 in Appendix B report results for another two parametrizations: $\alpha = 1{\cdot}1$ and $\alpha = 3$. Essentially, the policy effects are much larger for the former and much smaller for the latter. Table 4 in Appendix B reports results for another calibration strategy. There, instead of following Shimer (2005) in setting $\beta = \eta = 0{\cdot}72$, I keep $\eta = 0{\cdot}72$ and calibrate β jointly with $(\lambda, \varepsilon_0, \tau_b, \phi)$ so that the model also reproduces a labour share of 0·66. The general message from these exercises is that (i) the value of the curvature parameter of the Pareto distribution, α, is a critical determinant of the size of the effects that policy can have on TFP through composition effects (the effects tend to get larger as α approaches 1), and (ii) if factors are measured relatively inaccurately (*e.g.* when the TFP residual equals $\hat{A}$ or $\tilde{A}$), the policy effects on TFP are much larger and have the opposite sign than if the inputs are measured accurately (*e.g.* when the TFP residual equals A).

8. EXTENSIONS

In this section I extend the basic model to the case of serially correlated shocks and derive two additional aggregation results.

8.1. *Correlated shocks*

Section 4 established that, with some mismeasurement and equilibrium hoarding, the standard search model of the labour market with a particular structure of shocks generates a relationship between aggregate inputs and output that looks exactly like the standard Cobb–Douglas relation typically used in growth accounting exercises. The fixed cost $C(x, \phi)$ was introduced in Section 2 as a simple device to avoid "flat spots" in the value functions, and this made it possible for the equilibrium to exhibit hoarding.[38] Here, I show that by extending the model in a natural way, one can drop the fixed cost without affecting the main results. To this end, I generalize the productivity process by allowing for serially correlated shocks: when a match of productivity x suffers a change, the new value x' is a draw from the fixed distribution $G(x' \mid x)$. Assuming $G(x \mid x_1) < G(x \mid x_0)$ if $x_0 < x_1$ allows idiosyncratic shocks to be positively correlated through time. For this case, the cross section of productivities evolves according to

$$\frac{d}{dt}[(1-u_t)H_t(x)] = \lambda(1-u_t)\int_x^{\infty}[G(x \mid s) - G(R_t \mid s)]dH_t(s)$$
$$+ \theta q(\theta)u_t \int_{-\infty}^{\infty}[G(x \mid s) - G(R_t \mid s)]dH_t(s)$$

37. But note that financing the employment subsidy is relatively more costly than financing the hiring subsidy, since the former is paid out continuously to existing matches, whereas the latter just entails a one-time payment when the match is first formed.

38. For $C(x, \phi) = 0$, $\pi(x) = [\max(x - \phi, 0) - c]k - w(x)$, so $\pi(x)$ is flat up to ϕ and then rises with slope k. It is easy to show that in this case, $J(x)$ is also flat up to ϕ and then rises with slope $\frac{k}{r+\delta+\lambda}$. Note that since R is defined by $J(R) = 0$, this implies that generically the equilibrium with endogenous destruction will have $\phi < R$; that is, there is no hoarding except for the knife-edge case in which R is indeterminate. Ruling out these types of flat spots in J allows for the possibility that $R < \phi$ in an equilibrium with endogenous destruction. See Appendix B for details.

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$$- \lambda(1-u_t) \int_{-\infty}^{x} G(R_t \mid s) dH_t(s)$$

$$- \lambda(1-u_t) \int_{-\infty}^{x} [1-G(x \mid s)] dH_t(s) - \delta(1-u_t) H_t(x).$$

The first term accounts for the matches with productivities above x that get innovations below x but above R_t. The newly formed matches that start off with productivities no larger than x are in the second term. Notice the assumption that upon contact, the worker and firm draw their productivity level from the density corresponding to the average productivity among active matches.[39] The third term is the number of matches in the interval $[R_t, x]$ that get shocks below R_t and are destroyed. The fourth term accounts for the number of matches in the same interval that "move up" by virtue of having drawn productivities larger than x. The last term accounts for matches in the interval $[R_t, x]$ that are destroyed for exogenous reasons. Imposing steady states and rearranging yields

$$H(x) = \left[\frac{\lambda}{\delta+\lambda} + \frac{\theta q(\theta) u}{(\delta+\lambda)(1-u)} \right] \int [G(x \mid s) - G(R \mid s)] dH(s).$$

The steady-state unemployment rate is given by

$$u = \frac{\delta + \lambda \int G(R \mid s) dH(s)}{\delta + \lambda \int G(R \mid s) dH(s) + \theta q(\theta) \int [1 - G(R \mid s)] dH(s)}. \tag{29}$$

Using this expression, the steady-state cross-sectional productivity distribution becomes

$$H(x) = \frac{\int [G(x \mid s) - G(R \mid s)] dH(s)}{\int [1 - G(R \mid s)] dH(s)}, \tag{30}$$

a natural generalization of (3).

The problem of a searching firm is now summarized by

$$rV = -ck + q(\theta) \iint \max[J(z) - V, 0] dG(z \mid x) dH(x). \tag{31}$$

Again, entry of firms will occur until all rents are exhausted, so $rV = 0$ in equilibrium. The value of a filled job with productivity x is

$$rJ(x) = \pi(x) + \lambda \int \max[J(z) - V, 0] dG(z \mid x) - (\delta + \lambda)[J(x) - V], \tag{32}$$

where $\pi(x) = x \min(n, k) - \phi n - ck - w(x)$. Flow profit $\pi(x)$ is the residual remaining after the wage $w(x)$ and all other costs of production have been paid out. This formulation has only two such costs: the rental rate, ck, and the variable cost, ϕn.[40] The choice of hours that solves the

39. When shocks are independently and identically distributed, one can specify that new matches draw z from $G(z)$ just as active matches do when forced to update their shock. However, with correlated shocks active matches with state z draw the new shock z' from $G(z' \mid z)$. Since vacancies and unemployed workers have no productivity attached to them, I assume their initial draw z' is from the average density $\int G(z' \mid z) dH(z)$. As a way of motivating this, imagine—as do Mortensen and Pissarides (1994)—that firms must irreversibly adopt a "technology" to engage in production. The present specification then means that they draw their technology at random from all those active at the time the match is created. Jeffrey Campbell pointed out to me that Conlisk (1989) uses a similar assumption to determine the productivity of newly created plants in a model of technical change.

40. Here I model ϕn as a cost borne by the firm, but Appendix B shows that this formulation is equivalent to one in which ϕn is, instead, the disutility the worker experiences from working n hours.

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